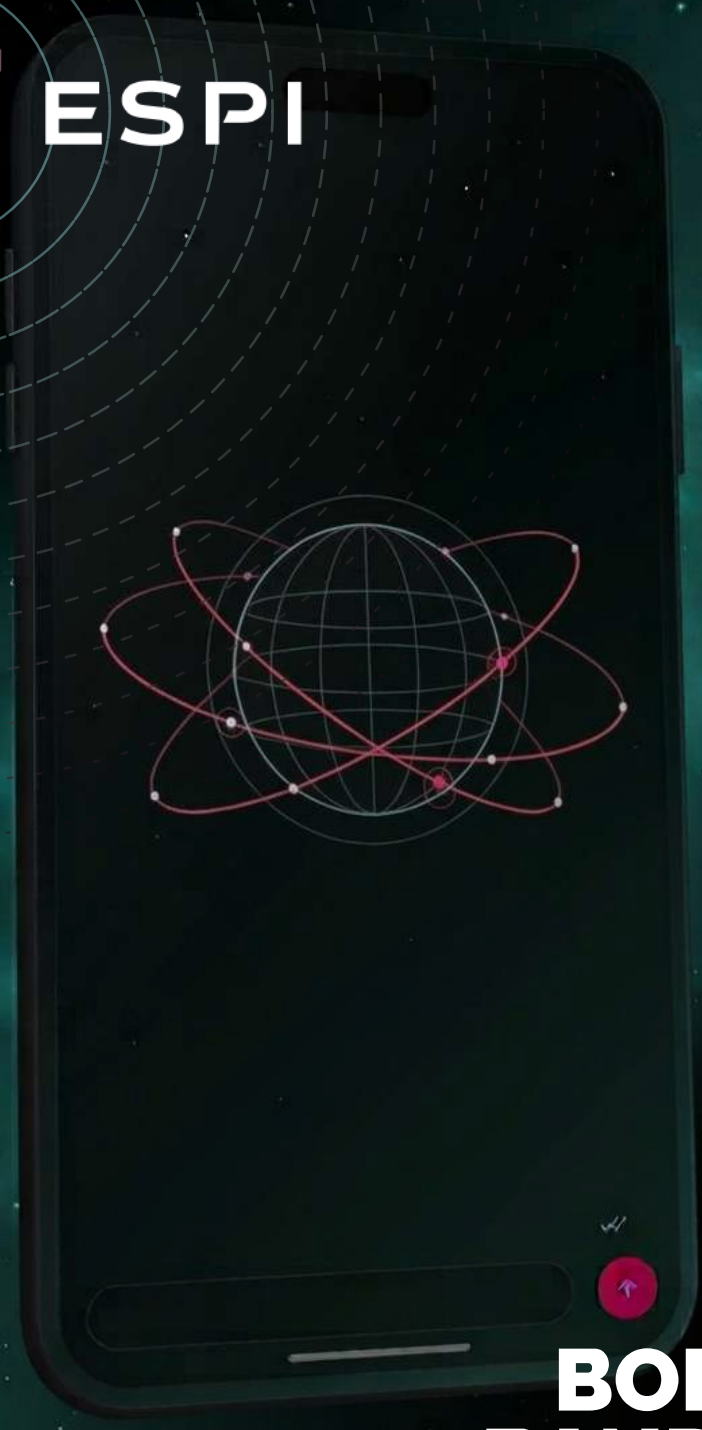
ESPI



BORROWED BANDWIDTH?

DIRECT-TO-DEVICE AND EUROPE'S AMBITION FOR FULL-STACK AUTONOMY

ESPI European Space Policy Institute

Brief No. 78

Borrowed bandwidth? Direct-to-Device and Europe's ambition for full-stack autonomy

Direct-to-Device (D2D) services are gaining traction across mobile communications market forecasts and business plans. Is Europe moving fast enough to secure its share? In May 2026, the **European Commission** proposed to reserve around two-thirds of the 2 GHz frequency band for mobile satellite services (MSS) - crucial for D2D - for European operators.[1] Meanwhile, SpaceX's **$19.6 billion acquisitions of EchoStar spectrum**, and Amazon's **pending $11.6 billion acquisition of Globalstar** demonstrate just how strategic and valuable spectrum has become and how speed is increasingly crucial to remain competitive.[2]

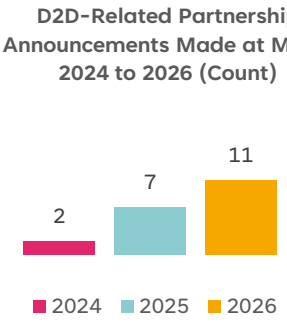


Figure 1: D2D-Related Public Announcements at MWC, 2024-2026

Significant challenges lie ahead for D2D to take off among European MNOs and satellite operators, symptomatic of broader underinvestment in space and digital sectors. This brief identifies these challenges and supports the design of European D2D policy pathways. For the purposes of this brief, D2D refers to the integration of **satellite connectivity into standard smartphones and consumer devices** without the need for specialised external hardware to access non-terrestrial networks (NTN).

D2D promises not only to fill gaps in ubiquitous connectivity across low-coverage areas and improve resilience in critical industries, but also fully redefine infrastructure economies for mobile communications. By 2034, the D2D satellite connectivity market is estimated to generate over $100 billion in cumulative revenues.[3] The D2D rollout therefore brings the potential to plug the space sector into a global mass market and offer a rare hyperscaling opportunity.

Mobile World Congress 2026 exemplified this fervour for D2D: there was a significant uptick in announcements of D2D partnerships between mobile network operators (MNOs) and satellite operators.[4] Also announced were innovations in underlying chipset architectures

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[1] European Commission, *Proposal for the EU-level authorisation of systems providing mobile satellite services (MSS),* May 2026 (Link); The EU's 2 GHz MSS spectrum was previously (2009) awarded to Inmarsat and Solaris Mobile, a JV between Eutelsat and SES. Following consolidation in the satcom landscape, these rights are now held by U.S. companies ViaSat and EchoStar respectively until May 2027.

[2] Harshita Mary Varghese, "Explainer: What is the low-band spectrum that SpaceX acquired for Starlink Mobile," Reuters, October 9, 2026 (Link); Reports on 8 October 2026 announced SpaceX agreeing to acquire Grain Management's nationwide 800 MHz spectrum portfolio reportedly at a price of $8 billion.

[3] Novaspace, *Prospects for Direct to Handheld and IoT Markets,* 10th Edition (Link).

[4] See Figure 1: methodology of landscape was conducted via filtered search of internet archives of past public announcements. Note: pre-2025/2026, D2D partnerships existed, but were not as widely publicised as today.

for devices connecting to satellites, signalling an ecosystem-wide push for D2D connectivity, and positioning satellite compatibility as the new baseline.

1 EUROPE'S CURRENT DIRECT-TO-DEVICE LANDSCAPE

Given D2D's rising importance, policymakers and businesses are trying to devise strategies boosting both sovereignty and competitiveness. Most notably, the European Commission has proposed to **reserve the majority of the 2 GHz MSS spectrum for European operators**. These efforts lay the foundation for autonomous European networks and would allow European operators to establish a baseline before turning to global markets.

Despite a strong legacy of satellite operators (e.g. Eutelsat Group, SES, Hispasat), native-European infrastructure for D2D services only exists at small scale or in early stages. OQ Technology and Sateliot operate their own IoT constellations, while SES is now investing in and partnering with U.S.-based Elveo with "*manufacturing capabilities and a robust supply chain in Europe*".[5] The D2D market in Europe is primarily underpinned by American satellite operators, namely Starlink and AST SpaceMobile, partnering with European MNOs, as illustrated in Figure 2.

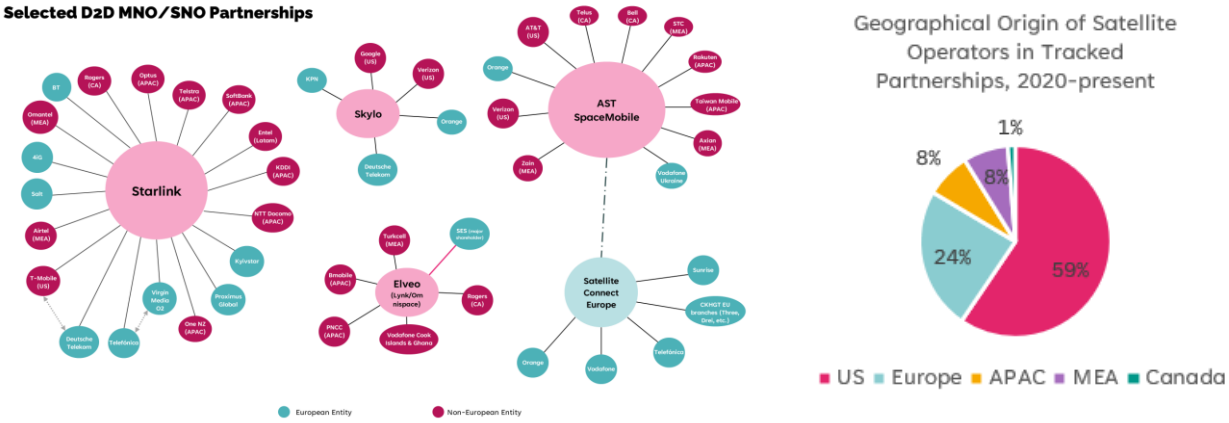


Figure 2: Selected D2D Partnerships Between Satellite Operators and MNOs; Geographical Distribution

Satellite Connect Europe (SCE) is a joint venture between AST SpaceMobile and Vodafone, further backed by Orange and Telefónica. It uses AST SpaceMobile's satellite infrastructure, whereby no data is processed on-satellite. Rather, data is simply "relayed" back to the MNOs' networks, which retain control over data processing within their ground-based networks (in contrast to Starlink, which processes data on-board). A similar solution is proposed by Equatys, a joint venture between U.S. Viasat and UAE Space42.

Sovereignty can be preserved at the level of data, but questions remain regarding control over the space infrastructure, and more generally, the trade-off between immediate

[5] SES, "SES Expands into Global Direct-to-Device Services through Strategic Collaboration with Elveo Mobile," press release, August 17, 2026 (Link).

capability access and long-term full-stack autonomy.

1.1 Out of Range: Why European D2D Has Yet to Take Off

While no single reason underpins Europe's relative lag in Direct-to-Device rollout and investment, there are systemic headwinds that must be understood before future-proofing policy action:

Uncertain ROI

Cautious European business mindsets generally limit appetite for higher-risk solutions such as D2D, **preventing the development of new strong industrial leaders at the same pace and scale of the U.S.**[6] European businesses see relevant use cases (e.g., remote enterprise IoT and emergency services) but view these as niche applications where market growth remains to be tested. In contrast, investments by U.S. satellite operators indicate D2D will become a cornerstone of their mass-market business models.

Technical debt

Without space-digital heavyweight partnerships, **Europe was left to rely solely on its legacy players to provide the risk & innovation pipeline.** But like their U.S. peers, European mature businesses are constrained by profit pressure and underinvestment in innovation, leading to technical debt and a cycle of financial strain. The result is that European capacity and technological building blocks required for D2D at-scale are lagging as operators are still catching up with the broadband connectivity revolution.

Disjointed incentives

This is compounded by **terrestrial connectivity subsidies that often distort immediate ROI** opportunities for satellite operators. In turn, these hinder the emergence of new innovation pipelines and the establishment of a space-digital stack in Europe. The EU, and its Member States, are spending billions in the deployment of fibre and terrestrial 5G infrastructures for underserved regions. Austria alone announced EUR 1.4 billion in fibre rollout in April 2021. Ultimately, this comes at the cost of interoperability and efficient roll-out of any native European LEO broadband and D2D solutions.[7]

Lack of a unified framework

Europe, including at national level, generally lacks a unified regulatory approach for satellite-terrestrial spectrum use akin to the U.S.'s Supplemental Coverage from Space (SCS) framework (with the exception of the UK).[8] Efforts on harmonisation are ongoing as the draft CEPT Report 93 was put out for consultation in June 2026.

Additionally, in Europe, spectrum has historically not been sufficiently prioritised as a **strategic resource**. **The Commission's proposal**, which reserves part of the 2 GHz MSS spectrum for European operators, **is therefore a welcome step**.

2 FROM SPECTRUM TO STACK: GAPS AND STRATEGIC NEXT STEPS

Beyond spectrum allocation, Europe needs to ask itself how 'European' a D2D service can reasonably be, and whether it wants its industry to play a role in a redefined global satellite

Shades of European

[6] S. Nixon, Europe's crippling risk aversion, Encompass, October 2024 (Link).
[7] European Commission, Digital connectivity in Austria, July 1, 2026 (Link).
[8] European Commission, *Commission Staff Working Document Executive summary of the impact assessment report Accompanying the document Proposal for a Regulation of the European Parliament and of the Council on digital networks*, SWD(2026) 14 final, January 2026 (Link).

connectivity market **across the full value chain**.

Partnerships such as Satellite Connect Europe or between 4iG and Starlink bring D2D to European customers at-pace. They, however, raise the policy question of whether operators relying on non-European space infrastructure should qualify for pre-reserved spectrum. Given Europe's ambition for digital sovereignty, anything but **maximising European control across the full value chain, including its space segment**, would be a suboptimal outcome. In line with Article 17 of the proposed MSS Regulation, safeguards to ensure the allocated spectrum remains under European control even in case of future M&A transactions should be enforced.

Spectrum with strings

Furthermore, there are **rumours of a joint bid** by a large group of European MNO incumbents. This could raise competition concerns if no safeguards are in place, leading to an outcome where highly prized spectrum is allocated to a group which already partners on D2D with a non-European satellite operator. Implementation should keep room for SME participation and remove entry barriers for European D2D solutions that come online in the coming years. This would **help prevent a quasi-sovereign outcome** where European spectrum and European brands unlock scale for U.S. satellite operators.

No early lock-in

This is **where IRIS² becomes relevant** – especially in its commercial dimension. The low-LEO sandbox, designed to test and enable advanced technologies, is crucial and should be given particular attention. If successfully run as a spiral development model, this can feed successive generations of IRIS² infrastructure while also derisking private investment into commercial European D2D services.

De-risking via IRIS²

Finally, **spectrum and supply-side interventions will not close these gaps alone**. Subsidies to close the digital divide remain largely fibre-centric. Making them technology-neutral would allow satellite solutions to openly compete for funding where they offer better value.

Tech-neutral rollout

Europe needs to prioritise a **full-stack approach**, particularly in technology areas with global mass-market potential. **In the case of D2D, this at minimum includes MNOs, satellite operators** and **spacecraft manufacturers**. Cracking one part of the equation, while ignoring the rest, will weaken Europe's aspirations for long-term digital sovereignty.[1]

Full-stack mindset

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